Venkata Lakshmi Parimala Pasupuleti

Founding Engineer · Agentic AI · LLM Evaluation & Validation · Production AI Observability · Data Pipelines · Secure Multi-Tenant Architecture · Python · AWS · FastAPI

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PROFESSIONAL SUMMARY

Engineer with 8 years owning complex, data-heavy systems end-to-end - and a track record of building the AI evaluation and validation infrastructure that makes production AI trustworthy. At Entrata, owned HUD compliance pipelines (fragmented federal data → 180,000 domain records), event-driven integrations for rent and eligibility workflows, and secure multi-tenant architecture serving 100K+ monthly active users at 99.5% uptime, reliability, and zero PHI security incidents. Built LeaseI, an 8-agent autonomous investigation platform (LangGraph + FastAPI) that diagnoses AI failure modes in housing compliance workflows with conditional HITL escalation. Built AgentProbe, an open-source production AI observability and evaluation toolkit: 5-metric eval engine, CI-integrated regression gate, LLM-as-judge scoring, and hallucination detection - the same system described in "build evaluation and validation systems around models and AI agents." M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Engineering & Backend: Python, FastAPI, PHP, JavaScript, TypeScript, SQL, Bash, Docker, Microservices, Distributed Systems, AWS (S3, EC2, RDS, Glue), Redis, RabbitMQ, REST APIs, Third-Party Integrations, Continuous Integration, CI/CD (GitHub Actions), PostgreSQL, MySQL, Oracle, Linux, Git

AI & Agents: Agentic AI, AI Agents, LLM Orchestration, LangGraph, LangChain, Multi-Agent Systems, RAG Pipelines, ChromaDB, LLM Evaluation, Validation, Failure Mode Analysis, Hallucination Detection, Agent Observability, Agent Tracing, HITL, Guardrails, Prompt Engineering, OpenAI API, Anthropic API, MCP, Production AI, FastAPI

ML & Analytics: Machine Learning, PyTorch, scikit-learn, NLP, BERT, spaCy, PySpark, Feature Engineering, Statistical Modeling, Tableau, Agile/Scrum

EXPERIENCE

Magizhs Tech Solution
May 2025 - Aug 2025

AI/ML Engineer Intern

- Built an agentic mission-planning system (LangGraph stateful graph) with end-to-end ownership: parse → plan → retrieve → validate → conditional routing - autonomous failure mode detection blocking non-compliant missions before export.
- Implemented RAG pipeline over FAA Part 107 corpus with validation guardrails enforcing altitude limits, battery budgets, and no-fly-zone checks; every compliance check grounded in cited source regulation.

George Mason University
Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built multi-source prediction models (Python, R) on WHO and UK Road Safety datasets - $R^2 = 0.977$ regression accuracy; applied Isolation Forest to automate data integrity checks across 3 research pipelines.
- Delivered Python and database workshops to 600+ students; designed Tableau and ggplot2 dashboards improving research comprehension by 40% across two cohorts.

Entrata - PropTech / Affordable Housing SaaS
Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Owned data ingestion pipelines end-to-end for 7 years** - fragmented federal source data expanded into ~180,000 domain records powering eligibility, rent, HAP, and tax-credit calculations for 100K+ housing units; designed validation and reconciliation at every stage; zero compliance violations across the ownership period.
- **Built event-driven RabbitMQ integrations** for gross-rent updates across thousands of tenant records - retry logic, idempotency guarantees, failure mode handling eliminating double-processing; owned through production incidents and annual rent-schedule changes.
- **Secure multi-tenant architecture with full data isolation** - led AWS S3 document migration for Affordable Housing module; achieved 100% PHI compliance with Libsodium-encrypted pipelines; zero security incidents across the entire resident data migration.
- **Shipped 20+ full-stack production modules** (Python, PHP, PostgreSQL, JavaScript) across 120+ Agile sprints; led Docker microservices redesign - 50% scalability improvement, 30% deployment reduction; eliminated 80% of manual operational workload through automation pipelines.
- Mentoring 5+ junior engineers; conducted 300+ peer code reviews; built observability tools reducing debugging time 40% and keeping bug count below 10 for 6+ consecutive months.

Entrata India - Pune
Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built property management modules (PHP, Python, PostgreSQL); led AWS S3 cloud migration resolving 90% of legacy storage constraints; income limits data ingestion covering 1.6M HUD/Tax Credit records across 5 years.

PROJECTS

AgentProbe: Production AI Evaluation & Observability Toolkit Eval · CI/CD

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) | github.com/venkatalakshmiparimala/AgentProbe

Built the evaluation and validation system described in this role - from scratch. Fully local, zero-cost AI agent evaluation toolkit: 15-endpoint FastAPI backend, 5-metric eval engine (correctness, groundedness, latency, token efficiency, hallucination rate), failure mode analysis, CI-integrated regression gate blocking regressions before merge, and multi-provider LLM routing for cost control. Used to validate LeaseI across 1,240+ compliance test cases with LLM-as-judge scoring and pass@k harness.

LeaseI: Agentic AI Investigation Platform (Housing Compliance) Agentic AI

Python · LangGraph (8-agent) · FastAPI · TF-IDF · OCR · SQLite | github.com/venkatalakshmiparimala/Lease_complaince_rag

8-agent autonomous investigation pipeline (Triage → Retrieval → Configuration → Historical Incident → Structured Data → Reasoning → Validation → Human Escalation) that diagnoses compliance failure modes across product settings, regulatory rules, and live system data - with conditional HITL escalation when confidence is too low. Achieved 93.6% precision, 87.9% F1 across 1,240 residential lease PDFs; CI-integrated regression gate with LLM-as-judge scoring.

AI on FHIR: NLP-Driven Medical Records Query System Healthcare

Python · Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings | github.com/venkatalakshmiparimala/AI_on_FHIR

Flask + React NLP platform translating natural-language queries into FHIR API integrations using LLMs, BERT, and embeddings; reduced manual clinical retrieval time ~60%; enforced HIPAA-compliant RBAC with secure data isolation across structured medical records.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017